

■ 3.3.3 Structural Operations on Rational Expressions

For ordinary polynomials, **Factor** and **Expand** give the most important forms. For rational expressions, there are many different forms that can be useful.

ExpandNumerator [<i>expr</i>]	expand numerators only
ExpandDenominator [<i>expr</i>]	expand denominators only
Expand [<i>expr</i>]	expand numerators, dividing the denominator into each term
ExpandAll [<i>expr</i>]	expand numerators and denominators completely

Different kinds of expansion for rational expressions.

Here is a rational expression.

$$\begin{aligned} \text{In}[1] := t &= (1+x)^2 / (1-x) + 3x^2 / (1+x)^2 + (2-x)^2 \\ \text{Out}[1] &= \frac{3x^2}{(1+x)^2} + \frac{(1+x)^2}{1-x} + (2-x)^2 \end{aligned}$$

ExpandNumerator writes the numerator of each term in expanded form.

$$\begin{aligned} \text{In}[2] := \text{ExpandNumerator}[t] \\ \text{Out}[2] &= 4 - 4x + x^2 + \frac{3x^2}{(1+x)^2} + \frac{1+2x+x^2}{1-x} \end{aligned}$$

Expand expands the numerator of each term, and divides all the terms by the appropriate denominators.

$$\begin{aligned} \text{In}[3] := \text{Expand}[t] \\ \text{Out}[3] &= 4 - 4x + x^2 + \frac{1}{1-x} + \frac{2x}{1-x} + \frac{x^2}{1-x} + \frac{3x^2}{(1+x)^2} \end{aligned}$$

ExpandDenominator expands out the denominator of each term.

$$\begin{aligned} \text{In}[4] := \text{ExpandDenominator}[t] \\ \text{Out}[4] &= \frac{(1+x)^2}{1-x} + (2-x)^2 + \frac{3x^2}{1+2x+x^2} \end{aligned}$$

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ExpandAll does all possible expansions in the numerator and denominator of each term.

In[5] := ExpandAll[t]

$$\text{Out}[5] = 4 - 4x + x^2 + \frac{1}{1-x} + \frac{2x}{1-x} + \frac{x^2}{1-x} + \frac{3x^2}{1+2x+x^2}$$

Together[expr]	combine all terms over a common denominator
Apart[expr]	write an expression as a sum of terms with simple denominators
Cancel[expr]	cancel common factors between numerators and denominators
Factor[expr]	perform a complete factoring

Structural operations on rational expressions.

Here is a rational expression.

In[6] := u = $\frac{(1+x)^2}{(1-x)(2-x)} + \frac{(1-x)^2}{(1+x)(2-x)}$

$$\text{Out}[6] = \frac{(1+x)^2}{(1-x)(2-x)} + \frac{(1-x)^2}{(1+x)(2-x)}$$

Together puts all terms over a common denominator.

In[7] := Together[u]

$$\text{Out}[7] = \frac{2 + 2x^2}{2 - 3x + x^2}$$

You can use **Factor** to factor the numerator and denominator of the resulting expression.

In[8] := Factor[%]

$$\text{Out}[8] = \frac{2(1+x^2)}{(-2+x)(-1+x)}$$

Apart writes the expression as a sum of terms, with each term having as simple a denominator as possible.

In[9] := Apart[u]

$$\text{Out}[9] = 2 + \frac{10}{-2+x} - \frac{4}{-1+x}$$

Cancel cancels any common factors between numerators and denominators. In this case, this leads to a simplification in the second term.

```
In[10]:= Cancel[u]
```

$$\text{Out}[10]= \frac{1-x}{2-x} + \frac{1+2x+x^2}{2-3x+x^2}$$

Factor first puts all terms over a common denominator, then factors the result.

```
In[11]:= Factor[%]
```

$$\text{Out}[11]= \frac{2(1+x^2)}{(-2+x)(-1+x)}$$

In mathematical terms, **Apart** decomposes a rational expression into “partial fractions”.

In expressions with several variables, you can use **Apart**[expr, var] to do partial fraction decompositions with respect to different variables.

Here is a rational expression in two variables.

```
In[12]:= v = (x^2+y^2)/(x + x y)
```

$$\text{Out}[12]= \frac{x^2 + y^2}{x + x y}$$

This gives the partial fraction decomposition with respect to x.

```
In[13]:= Apart[v, x]
```

$$\text{Out}[13]= \frac{y^2}{x(1+y)} + \frac{x(1+y)}{1+2y+y^2}$$

Here is the partial fraction decomposition with respect to y.

```
In[14]:= Apart[v, y]
```

$$\text{Out}[14]= -\left(\frac{1}{x}\right) + \frac{y}{x} + \frac{1+x^2}{x(1+y)}$$